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Ph.D. Probability. May 2003

WRITE CLEARLY. STATE PRECISELY
RESULTS YOU USE. DO NOT
OMIT STEPS.

ANSWER EACH QUESTION
ON A SEPARATE SHEET.

1. Let $(X_n)_{n=1}^{\infty}$ be an i.i.d.
sequence of real valued variables
on $(\Omega, \mathcal{F}, P)$. Let $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$.
Put $S_n = X_1 + \dots + X_n$, for $n \geq 1$ and
put $S_0 = 0$.

Which of the following sequences
are martingales relative to $\mathcal{F}_n$. For those
which are not martingales w.r.t. this
filtration, what conditions have to be

1. S_n 2. S_n^2 3. $S_n^2 - n$ 4. $S_n^2 - n^2$ 5. $S_n^2 - n^3$

(2)

a) $S_n, n=0,1,2,\dots, E[|X_1|] < \infty$

b) $\bar{X}_1^2 + \dots + \bar{X}_n^2 = n\lambda, n=1,2,\dots, E(X_1^2) < \infty,$

λ real

c) $\exp(\alpha S_n - n\lambda), n=0,1,2,\dots$ where

$\varphi(\alpha) = E[\exp \alpha X_1] < \infty, \alpha, \lambda$ real.

d) $T_n = S_{\min(n, T)}, n=0,1,2,\dots$

where

$P(X_1 = \pm 1) = \frac{1}{2}$ and $T = \min(n > 0, S_n = 0)$

[2] Let $\{\alpha_n\}_{n=1,2,\dots}$ be a sequence of probability measures on $\mathbb{R}$. Show the following are equivalent:

a) There exists a probability measure α on $\mathbb{R}$ such that

$$\lim_{n \rightarrow \infty} \alpha_n(I) = \alpha(I)$$

(3)

for all closed intervals I on $\mathbb{R}$ whose end points are continuity points of α .

b) If $\{F_n\}_{n=1,2,\dots}$, respectively F are the distribution functions of $\{\alpha_n\}$, respectively α ,

$$\lim_{n \rightarrow \infty} F_n(x) = F(x)$$

at every point x of continuity of F .

[3] Let $\{\alpha_n\}_{n=1,2,\dots}$ be a sequence of probability measures on $\mathbb{R}$ and let $\{\phi_n(t)\}_{n=1,2,\dots}$ be the corresponding characteristic functions.

Assume that $\lim_{n \rightarrow \infty} \phi_n(t) = \phi(t)$ exists.

$\forall t \in \mathbb{R}$ and $\phi(t)$ is continuous at $t=0$.

Prove that $\phi(t)$ is the characteristic function of some probability measure and that the conditions of Problem [2] hold.

(4)

[4] If X is a real valued random variable,
show that for $0 < \epsilon \leq 1$,

$$E\left[\frac{|X|}{1+|X|}\right] \leq \epsilon + P[|X| > \epsilon]$$

and that

$$P(|X| > \epsilon) \leq \frac{1+\epsilon}{\epsilon} E\left[\frac{|X|}{1+|X|}\right].$$

Deduce that the set of real valued
random variables on a given probability
space can (with suitable identification)
be regarded as a metric space if the
distance between two random variables
 X and Y is defined to be

$$d(X, Y) = E\left[\frac{|X-Y|}{1+|X-Y|}\right].$$

and that convergence in this metric space
then corresponds to convergence in probability.

(5)

[5] a) Let $\underline{X}_n, n=1,2,\dots$ be i.i.d >0 ,
 $E[\underline{X}_n] = 1$ and $\underline{Y}_n = \prod_{k=1}^n \underline{X}_k$. Prove that
 $\underline{Y}_n$ is a martingale relative to the
 filtration $\mathcal{F}_n = \sigma(\underline{X}_1, \dots, \underline{X}_n)$.

Show that $\lim \underline{Y}_n = \underline{Y}_\infty$ exists a.s.

b) Show that

$$P[\underline{Y}_\infty = 0] = 0 \text{ or } 1.$$

Hint: Recall infinite products and
 the zero-one law.

c) (Using b)) Show that

$$E[\underline{Y}_\infty] = 1 \iff P[\underline{Y}_\infty > 0] = 1.$$

[6] Let $\underline{X}_n, n=1,2,\dots$ be a supermartingale
 such that $E[|\underline{X}_{n+1} - \underline{X}_n| | \mathcal{F}_n] \leq M$ a.s.

Show that for any two optional times $S \leq T$,
 $E[T] < \infty$, we have $E[|\underline{X}_T|] < \infty$ and
 $E[\underline{X}_T | \mathcal{F}_S] \leq \underline{X}_S$.